# Incremental Graph Layout by Stochastic Search: a Routed Objective, a Calibrated Metric, and a Matched-Budget Control

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**Abstract**

Placing the nodes of a graph in the plane with few edge crossings is NP-complete, so every layout tool runs heuristics, and the heuristics are rarely explicit about three things: what objective they optimize, whether that objective measures the drawing the reader actually sees, and whether the search spends its budget any better than uniform random sampling would. We treat all three. The problem is formulated incrementally: a graph grows by $k$ nodes and the existing drawing is frozen, which keeps the reader's mental map, bounds the search space at $2k$ variables, and contains the classic full redraw as the limit of an anchor term that prices the displacement of frozen nodes. The objective reads the drawing's real medium, orthogonally routed connectors rather than the straight center-to-center segments most objectives score, and the metric is calibrated: the router's shortfall against a reference layout of certified on-screen quality is measured and printed beside every number that depends on it. Four families of stochastic search (hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform random sampling as the matched-budget control) are compared by exact sign tests on paired seed panels, deterministically reproducible to the byte across platforms. The motivating application is the entity–relationship diagram, which mainstream tools still place badly enough that maintainers redraw by hand; a proof of concept places the tables added by eight real consecutive schema migrations of a production database, each with a human-accepted answer, under a pre-registered analysis. The genetic algorithm and hill climbing each beat a centroid placement heuristic on all eight migrations (exact Wilcoxon, Holm-corrected $p = 0.023$), though so does the control, so those comparisons measure budget against no budget; the comparison with teeth is against the control, and there the pre-registered refutation clause fired: the genetic algorithm did not separate from uniform sampling. Hill climbing, not the genetic algorithm the author predicted, is the budget-efficient method, directionally ahead of it on every non-tied migration and separating from the control on seven of eight pairs, six of them by 2000 evaluations; on single-table migrations every method's margin over sampling is under three tenths of a percent, so the first deployable rule is to count the new nodes before reaching for a search. One general argument: under this objective the searches outscore the human reference, which measures the objective's blindness to what maintainers optimize rather than any superiority of the tool, and we argue this gap is why automatic graph layout has stayed bad in practice.

## 1 Introduction

For most of two decades the author maintained the entity–relationship diagram of a production database by hand. Every reverse-engineering of the schema replaced the diagram's positions with the tool's own placement, and restoring a 44-table diagram to readability cost about an hour each time. This is not a defect of one tool. MySQL Workbench's placement algorithm is undocumented (its manual names only the menu item), and its failure mode is on the record:

*Independent researcher. Code, data and derived tables: `https://github.com/Anode1/cjitter`.

bug #42600 reports a 353-table schema autoplaced with some ninety tables stacked into one pile. Commercial tools exist that draw entity–relationship diagrams with straight diagonal edges, a representation no maintainer draws by hand.

The underlying task is hard in the formal sense: placing the nodes of a graph in the plane with the fewest edge crossings is NP-complete [1], so every tool runs heuristics. Our claim is that the practical failure is not the hardness but three things the heuristics leave unstated. First, the objective: what quantity, exactly, is being improved. Second, the medium: whether that quantity is measured on the drawing the reader sees. A tool that optimizes straight center-to-center segments while rendering orthogonal connectors improves a drawing nobody is shown. Third, and the one no study in graph layout runs: the control. Label placement, the adjacent field this work descends from, included a random baseline in its landmark evaluation [7], and matched-budget random controls are standard doctrine in hyperparameter search [18]; in graph layout we have not found a study that runs the control at matched budget, and without it the claim "our method improves layout" is unfalsifiable at any fixed cost.

This paper treats all three, on the practical version of the problem, which is incremental: a migration adds $k$ tables to a diagram whose other tables must not move, because a reader who knows where a table sits should still find it there. Section 2 formulates the problem and its one-parameter generalization to the classic full redraw. Section 3 makes the objective read the drawing's real medium and calibrates the metric against the one certified fact available; Section 4 defines the metrics and the statistics. Sections 5 and 6 describe the search harness and a benchmark of real consecutive migrations with human-accepted answers. Section 7 reports a pre-registered experiment whose refutation clause fired against its own author's prediction, which we regard as the methodology working; Section 8 examines what beating the human reference does and does not mean.

## 2 The problem

Let $G$ be a graph with a drawing $D$ that assigns each node an axis-aligned rectangle in a bounded canvas, and let $G'$ be a supergraph adding $k$ nodes and their edges. The *incremental placement problem* is to position the $k$ new rectangles, with $D$ frozen, minimizing an objective over the resulting drawing, subject to two hard constraints enforced by construction: no two rectangles overlap, and every rectangle lies on the canvas. The search space is $2k$ real variables however large $G$ is.

The frozen context is not a concession to tractability. The dynamic-graph-drawing literature names the reason the *mental map* [3]: a reader who has learned a drawing owns its geography, and a layout that moves what the reader knows destroys more value than it adds. The empirical evidence on mental-map preservation is mixed, helping some reading tasks and not others [15]; the migration economy is the sharper argument here, since the maintainer's acceptance is the task. Each migration adds a few tables, the diagram's owner accepts a placement, and the accepted drawing becomes the frozen context of the next migration. A version-controlled schema therefore accumulates ground truth: every revision pair is an instance, and the human's accepted answer for it.

One parameter connects this problem to the classic one. Add to the objective an anchor term $\lambda \sum_i \|p_i - p_i^0\|^2$ over the frozen nodes' displacements from their accepted positions and let them move: $\lambda \to \infty$ recovers the incremental problem, $\lambda = 0$ the full redraw, and intermediate values price the reader's geography in the objective's own units. This paper measures the $\lambda \to \infty$ end, where the practical need is; the continuum is the natural sequel, with thirty years of force-directed practice [17] as the control at the unfrozen end.

## 3 The routed objective

Entity–relationship tools draw orthogonal connectors: polylines at 0 and 90 degrees, leaving a table's border, bending once or twice. An objective over straight center-to-center segments scores penetrations and crossings that do not exist on screen and misses ones that do. We therefore route before we measure. Each edge is routed as the cheapest of 24 orthogonal candidates (two L shapes and Z shapes whose middle segment slides across the channel between the endpoints and steps beyond it), anchored on the facing borders at an attachment point unique to that edge, so no two connectors ever share a segment. Routing is sequential and aware: an edge's candidates pay for crossings against every connector already placed, so each pair of edges is priced exactly once and a route can dodge a connector instead of crossing it. The choice is deterministic, ties resolved by declaration order.

The objective is then $S = 100P + 100C + L$ over the routed drawing: penetration $P$ as the total length of connector lying inside tables it does not join (a length, never a count, because a count is flat under small moves and gives a search nothing to follow), crossings $C$ as a count at the same tier, and routed length $L$ two orders of magnitude below. Node overlap and canvas containment are not terms: they are enforced by a repair callback before any candidate is scored, so no infeasible drawing can be returned as anyone's best. Optimizing the wrong medium has a measurable price: on the pilot pair, a layout optimized under the straight model retains 830 units of routed penetration where the routed optimizer reaches zero.

Orthogonal connector routing is solved to optimality in deployed libraries [10]; ours is deliberately cheap, two bends and a fixed candidate set, because it runs inside the objective at every evaluation, and the price of that choice is measured rather than assumed. A routed metric is an approximation of the tool's rendering, so we calibrate it against the one certified fact available: the maintainer's accepted layout achieved zero crossings and zero edges under tables on screen. Our router, applied to that layout, measures 27 crossings and 323 units of penetration; the shortfall, largely the gap between a two-bend router and an optimal one, is printed beside every number that depends on it, and no comparison against the human reference is read past it. The calibration also disciplined development: each router improvement (border anchors, attachment slots, crossing-aware sequencing) was accepted because this number fell, from 36 crossings and 491 penetration at the first orthogonal router to the present values. The same drawing under the straight-segment model charges the certified layout 20 crossings and 1944 penetration, which quantifies how far the diagonal representation misdescribes what the maintainer made.

## 4 Metrics and statistics

The score is $S$ of Section 3, computed identically for the straight model by substituting the direct segment for the route. Beside $S$:

- the *feasibility pair* $(C, P)$, ordered lexicographically: the part of the score a reader can verify by looking at the drawing;
- the *calibration* $\chi = (C_{\text{ref}}, P_{\text{ref}})$, the metric applied to a reference layout whose on-screen quality is certified as zero and zero; the gap between $\chi$ and $(0, 0)$ is the metric's own error and caps every reading against the reference;
- the *transfer gap* $\Delta = S_{\text{routed}}(\hat{v}_{\text{straight}}) - S_{\text{routed}}(\hat{v}_{\text{routed}})$, the routed cost of having optimized the straight model (the pilot's penetration version appears in Section 3);
- the *separation budget* $B^*$: the smallest evaluation budget at which the method beats the control on every one of the five shared seeds, the one-sided sign test's only route to $p \leq 0.05$ at that panel size. $B^*$ is descriptive: it detects sweeps (probability $q^5$ at per-seed win rate

$q$), is censored to the tested budget grid, and a finite value can rest on tie-breaker margins, so $\infty$ marks non-detection at that power, never equivalence.

Within an instance, methods share one seed panel and are judged by the exact one-sided sign test on paired per-seed differences. Across instances, the pre-registered comparisons use the exact Wilcoxon signed-rank test with Holm correction over the declared family, exact McNemar on the solve-count binary, Hodges–Lehmann shifts with distribution-free intervals, and a minimum detectable effect wherever a null is reported, so a null bounds rather than asserts. Realized: $n = 8$ pairs (Section 6), five seeds per method per pair.

# 5 The searches and the control

The harness, cjitter,[1] runs four methods behind one fitness interface: hill climbing with restarts, simulated annealing, a genetic algorithm with tournament selection and blend crossover, and uniform random sampling as the matched-budget control. Every evaluation passes through one counter, so no method can spend more than another; the budget is exact by construction, and an internal guard turns any violation into an immediate failure rather than a silently unfair comparison. All four run on the same seed panel, and the verdict against the control is an exact one-sided sign test on the paired per-seed differences: "better" requires the wins to be explainable by a fair coin with probability at most five percent, and anything else prints as "not shown," a failure to demonstrate, not evidence of equality.

Two design choices carry the paper. First, determinism: every trajectory uses integer arithmetic and the two libm calls IEEE 754 computes exactly or rounds correctly (fabs, sqrt); the gaussian step is a sum of uniforms rather than Box–Muller, the annealing schedule uses an arithmetic exponential, and a run reproduces byte for byte across compilers and platforms, which is what lets this paper pin every reported number as a regression test. Second, the control: the first genetic algorithm shipped in this harness mutated at a fixed scale, re-scattering whatever the population had converged to, and the control's verdict read "no better than uniform sampling at equal cost." An outside review traced the verdict to the fixed scale; with mutation decaying over the run, the same algorithm became the best method on the harness's label-placement example (90 rectangles in a container, minimum overlap). A method-versus-method table without a random control cannot distinguish a weak method from a mis-tuned one.

# 6 The benchmark

The instances are real. One production schema's version history holds seventeen revisions of its MySQL Workbench model, sixteen consecutive pairs, each a migration whose diagram a maintainer restored by hand and accepted on both sides. Per pair, the task is the previous revision's diagram frozen as drawn, the added tables to place, and the maintainer's accepted coordinates in the next revision as the human reference; the median displacement of surviving tables between revisions, 36 to 1254 canvas units across the usable pairs, is recorded per pair, since the human answered in a context that sometimes moved. The pre-registered low-displacement subset (at most 20 units) is empty, so that confound on the human reference is uncontrolled here. Seven of the sixteen pairs added no tables (documentation edits, pure re-layouts) and are excluded with that reason; one pair served as the pilot and is excluded from the primary analysis; eight remain. Two threats are recorded rather than repaired: matching is by table name, so a migration that renames tables can count renames as additions (pair 14, "refactoring and renaming," is the flagged case); and the eight pairs are consecutive migrations of one schema, each frozen context containing the previous pair's answer, where the exact tests assume independent pairs, so generalization beyond this database is external validity, not statistics.

[1] C99, no dependencies beyond libm; the repository carries the suites, the data and every number in this paper.

Figure 1: One benchmark instance, anonymized. Left: the frozen diagram before the migration (36 tables). Right: after it, the ten added tables in amber, connectors routed orthogonally. The maintainer's accepted coordinates for the added tables are the human reference.

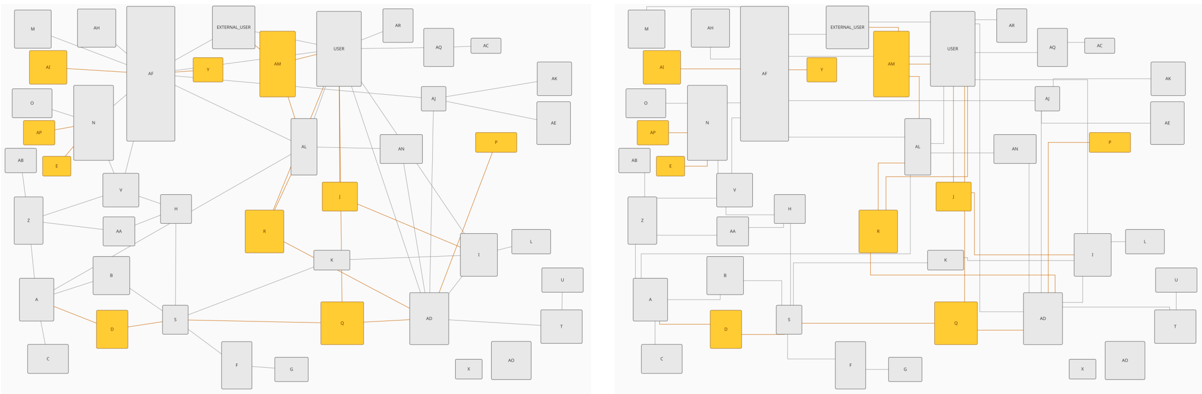

Figure 2: The same drawing under the two edge models: straight center-to-center segments (left) and routed orthogonal connectors (right), what the maintainer's tool draws and this paper's objective reads.

The current pair ships with the harness as its example, anonymized for publication: every table renamed to a letter, business column names neutralized down into the view SQL, the machine identifier that v1-style UUIDs embed scrubbed, and the geometry, which is the ground truth, untouched. The mapping back to the schema is deliberately stored nowhere.

## 7 Pre-registered experiment

The analysis was committed, timestamped, before any non-pilot pair was scored: primary comparison, declared families, refutation criterion, and the author's own directional prediction. Per pair and method, five seeds at the shipped budget of 8000 evaluations (with 500, 2000 and 32000 for the separation-budget table), the per-pair value the median over seeds; the statistics of Section 4 throughout.

The primary comparison succeeded, and we state at once how far it reaches. The genetic algorithm beats the centroid heuristic (each new table at its neighbours' centroid) on eight of eight migrations, exact Wilcoxon $p = 0.0078$, Holm-corrected 0.023, Hodges–Lehmann shift $-72{,}287$ $[-150{,}515, -20{,}728]$; hill climbing reads the same; and so does uniform random sampling, which also beats the centroid on all eight, so these comparisons measure a budget of evaluations against none. At $n = 8$ the exact test's 0.0078 is its smallest attainable value, an eight-for-eight sweep, so the Holm-corrected 0.023 carries no margin. On the pre-registered feasibility order $(C, P)$

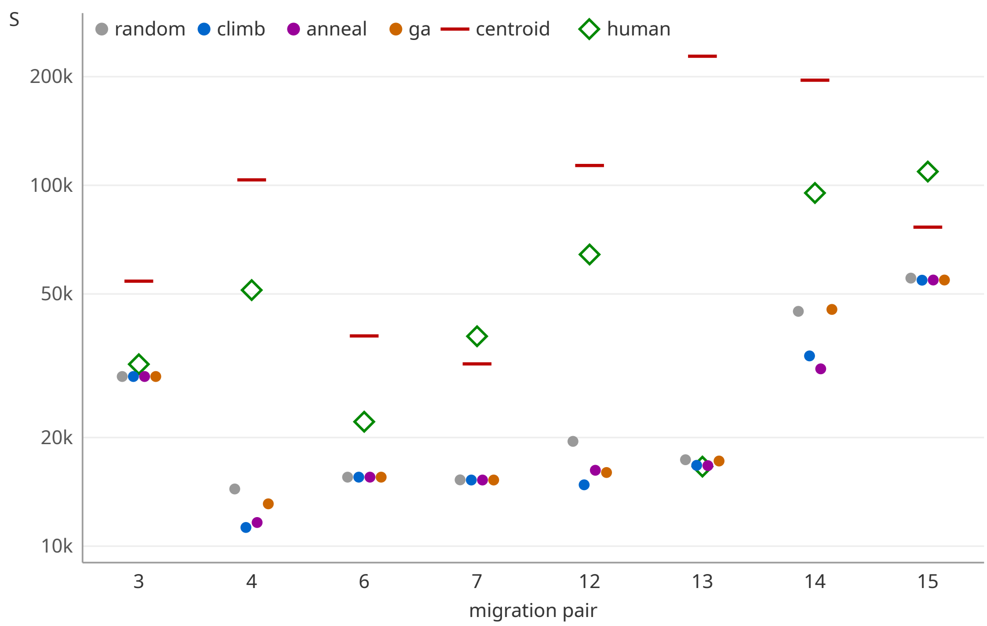


Figure 3: Per-pair medians of the routed objective at budget 8000 (five seeds), log scale, with the centroid heuristic and the human reference. On the single-table pairs the four methods nearly coincide; the heuristic loses everywhere on $S$ (on the feasibility order the comparison reverses; Section 7); the human reference carries the router's calibration error and is a reference, never a target.

Table 1: Separation budget $B^*$ per pair: the smallest budget (of 500, 2000, 8000, 32000) at which the five-seed sign test against the matched-budget control reaches $p \leq 0.05$; $\infty$ means not detected at any tested budget. The median column ranks $\infty$ above every finite budget; the never column counts the $\infty$ entries.

| pair | 3 | 4 | 6 | 7 | 12 | 13 | 14 | 15 | median | never |
|---|---|---|---|---|---|---|---|---|---|---|
| climb | $\infty$ | 2000 | 2000 | 2000 | 2000 | 8000 | 2000 | 500 | 2000 | 1 |
| anneal | $\infty$ | 2000 | $\infty$ | 32000 | 500 | 8000 | 500 | 500 | 5000 | 2 |
| ga | $\infty$ | 32000 | 2000 | 500 | 500 | $\infty$ | $\infty$ | 8000 | 20000 | 3 |

the direction reverses: the centroid beats the GA on four pairs against one (three ties) and hill climbing three against two, the searches buying zero penetration at the price of crossings; neither direction approaches significance, and Figure 3's "loses everywhere" is a statement about $S$ alone. The Friedman omnibus over the four methods rejects at $p \approx 0.0005$ with mean ranks climb 1.44, anneal 2.19, GA 2.69, random 3.69; the Nemenyi post-hoc promised in the pre-registration is omitted as redundant to the reported pairwise family, a declared deviation.

Two declared clauses then fired against the experimenters, which we report with the same prominence. The refutation clause: the GA did not separate from uniform random sampling. The decomposition matters: six wins, one tie (the fully degenerate pair 3), and one outright loss to the control on pair 14, the rename-flagged pair, for exact Wilcoxon $p = 0.109$; flipping that single pair would give $p = 0.016$, so the verdict is carried by one loss on a suspect instance and by $n$. The easy pairs are real but they are not the whole story: on the three single-table migrations every method's margin over the control is at most three tenths of a percent (pair 3 is identical to five digits), while the one $k = 2$ pair shows a genuine 3.5 percent search win, so the deployable rule is stated for $k = 1$ and by effect size: count the new nodes before reaching for a search.

And the author's pre-registered prediction, that the GA would be the most efficient search, was refuted by its own experiment. Hill climbing is directionally ahead of the GA on every non-tied migration (seven of seven, exact $p = 0.016$, Holm 0.031); the tally's robustness has a limit we state rather than hide: two of those wins are margins of 0.02 and 1.6 score units on the near-degenerate pairs, below the within-pair seed spread of about 27, and treating them as ties gives five of five, $p = 0.0625$, so the ordering is directionally consistent and not robust to a tie threshold the pre-registration failed to declare. The declared family's other half is reported with it: GA versus annealing is not shown ($p = 0.297$). Climbing separates from the control on seven of eight pairs, six by 2000 evaluations (the GA: five of eight, three by 2000; medians in Table 1); an exploratory bound finds climbing and annealing indistinguishable to within a tenth of the primary effect. The pilot pair, with $k = 10$, had pointed the other way, GA first; the surviving hypothesis, that recombination pays only when there are enough modules to recombine, is the successor pre-registration, and it will be scored the same way this one was.

# 8 What the human comparison measures

Under the routed objective the searches outscore the maintainer's accepted layouts on most pairs. Read through the calibration line, this is not a claim that the tool out-draws a person. Much of the human reference's score is the router failing to reproduce connectors the maintainer's tool routed cleanly; and the feasibility pairs disagree with the scores, the searches reaching zero penetration while the human's on-screen crossings stay lower. Person and tool jointly achieved zero and zero; each automated party wins a different half of that, and the router owns the remaining gap.

A layout objective of this family sees crossings, penetration and length; a maintainer optimizes semantic grouping, aligned rows and room to grow. Any such objective will therefore beat the human on itself and lose on the screen, and a tool guided only by such an objective ships the drawing the reader then spends an hour repairing. The gap between drawing metrics and human judgment has an empirical literature of its own [14]; what the migration benchmark adds is the maintainer's acceptance as the recorded outcome. That gap, between what layout objectives score and what maintainers accept, is our explanation for why automatic diagram layout has stayed bad in practice despite methods that demonstrably beat their controls: the searches are fine; the objectives do not yet describe the job. Objectives anchored to certified human layouts, with their approximation error measured and printed, are the repair this paper proposes.

# 9 Related work

Crossing minimization is NP-complete [1]; orthogonal drawing with bend minimization is a literature of its own [4], founded industrially on exactly our application, the orthogonal drawing of database diagrams [5]. Stochastic search for layout begins in earnest with Davidson and Harel's simulated annealing over aesthetic criteria [2] and continues through genetic approaches such as TimGA [6]; label placement ran the landmark empirical comparison, random baseline included [7]. The mental map was named by Misue, Eades, Lai and Sugiyama [3], dynamic graph drawing is surveyed in [16], and geometric incremental placement has direct precedents: interactive orthogonal drawing with no-change insertion scenarios [8], and Dunnart's constrained interactive layout with routed connectors and a preserved drawing [9]. Orthogonal connector routing is solved to optimality in deployed libraries [10]. On the layered side, the Constrained Incremental Graph Drawing Problem carries the metaheuristics line [11, 12], most recently hybridized with graph representation learning [13], over synthetic instance sets. Force-directed practice is surveyed in [17]; graphviz's `neato`, honoring pinned positions, is the baseline the anchor-term sequel must face. The empirical gap between drawing metrics and human judgment is Purchase's line [14].

Against that background the novelty claim is narrow and we state it narrowly. Interactive tools place into frozen drawings and route connectors [8, 9]; evaluations have included random baselines [7]; each ingredient exists somewhere. What we have not found together is the evaluation half: an objective calibrated against a certified human layout with its error printed, a random control at matched budget in graph layout, exact paired inference over real migration instances with human answers, and a pre-registration whose refutation clauses were allowed to fire, one of them against the author.

## 10 Conclusion

An hour of hand repair per reverse-engineering, for years, was a measurement nobody wrote down. Written down properly, it became: a formulation (frozen context, $2k$ variables, an anchor term to the classic problem), an objective that reads routed connectors with its calibration error measured and printed, a harness in which four searches spend identical budgets against a random control and are judged by exact paired tests, a benchmark of eight real migrations with human-accepted answers, and a pre-registered experiment that ranked the searches above the centroid heuristic, refuted the author's own prediction, and produced one deployable rule, from eight migrations of one schema and stated with that scope: count the new nodes first. The successor questions are declared and waiting: whether recombination's advantage returns with the module count, whether the per-label climber that started this line in 2001 holds its own inside the harness that finally names it, and at what objective noise any search stops being worth its budget.

## Reproducibility

Every number in this paper regenerates from the repository at the frozen commit: the per-seed data, the extraction and analysis code, the pre-registration with its addenda and this paper's source live together, and the harness's own test suite pins the shipped example's every printed digit. The benchmark geometry derives from a private schema and is carried anonymized; the anonymization mapping is stored nowhere.

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